

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 1st Year 1st Term Examination, 2014
URP 1111
(History/Evolution of Human Settlement)

Full Marks 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define human settlement. List down the elements of human settlement with examples in tabular format. (08)
- (b) What do you understand by the term 'Population Density'? Explain the third and fourth principles of human settlement. (10)
- (c) Sketch the parts of human settlement in case of urban form. (05)
- (d) Explain Static Urban Settlement with example and provide necessary figure. (12)
2. (a) 'Further you move from the city center, buildings become lower'-justify the concept. (05)
- (b) Write in brief the housing pattern in Neolithic period and provide sketch, if necessary. (10)
- (c) Why are the middle slopes preferred for a farm house? (05)
- (d) Write short notes (any three): (15)
(i) Primitive organized settlement (ii) Gossip square (iii) Star-like pattern
(iv) Primate city (v) The ring city
3. (a) Briefly describe the origin of garden city and provide necessary sketches. (15)
- (b) Provide your understanding about the concept of vertical city of Le Corbusier. (20)
4. (a) What was the fate of the Radburn city in New Jersey? (05)
- (b) Briefly explain the fundamental elements of neighborhood concept. (15)
- (c) Write short notes (any three): (15)
(i) Pundravarddhan (ii) Vanga (iii) Samatata (iv) Harikela (v) Shat Gumbaj Mosque

Section – B

5. (a) Mesopotamian Civilization was known as the "Cradle of Civilization". What are the main features behind this statement? (10)
- (b) What was the city planning concept during the Mesopotamian Civilization? (10)
- (c) Why was the Hammurabi period referred to as 'Golden age of Babylon'? (05)
- (d) 'At early settlements there were no planning regulations, people built houses here and there'-Justify according to initial concepts of settlement patterns. (10)
6. (a) Describe the town settlements during Egyptian Civilization where 'Set Back Rules' were first introduced. (10)
- (b) 'Pyramid' is the unique sign and master piece of Egyptian Civilization –what was the secret concept behind this monument? (10)

- (c) Write short notes on any two (02) of the following issues. (15)
(i) Golden age of Babylon (ii) Athenian democracy (iii) Serfs at Medieval period
7. (a) What do you understand by 'period of chaos and destruction'? (05)
(b) Write an overview of the greatest amphitheatre and famous landmark in Roman period with necessary diagrams. (10)
(c) Describe briefly 'Mohenzodaro' and 'Harappa' city planning concept during Indus Valley Civilization. (12)
(d) How was the housing condition during Minoan Civilization? (08)
8. (a) Why 'Parthenon' is an international symbol at present time? Which order was followed in Parthenon during Greek period? Describe with necessary sketches. (15)
(b) What was the principle concept of grid iron street pattern in Greek period? Draw the pattern. (05)
(c) Why 'Colosseum' was the most famous land mark in Roman period? Elaborate with necessary diagrams. (08)
(d) How was the public water supply system during Roman period? (07)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2014
HUM 1117
(Functional English)

Full Marks 210

Time: 3 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make appropriate WH questions from the following: (14)
 - i) Her politeness pleased us.
 - ii) Now it is 11 a.m.
 - iii) The weather is stormy.
 - iv) Everybody wishes to be happy.
 - v) He earns 20 thousand per month.
 - vi) Bangla is our mother tongue.
 - vii) He talked gaily.
- (b) Make sentences with each of the following words as directed. (12)

Record (as Noun); Record (as Verb); Record (as Adjective); Still (as Noun); Still (as Verb); Still (as Adjective).
- (c) Change the following words as directed and use them in sentences. (09)

Force (into Verb); Book (into Adjective); Poor (into Adjective); Blood (into Verb); Habit (into Verb); Habit (into Adjective).
2. (a) Make new words with each of the following prefixes and suffixes and use them in sentences. (14)

Ab..., Co..., Mal..., Bi..., ...let, ...en; ...age.
- (b) Make one antonym and one synonym of each of the following words and use them in sentences. (12)

Reveal; Strict; Cheerful; Liberty.
- (c) Make sentences using the following phrases and idioms. (09)

Hard and fast; End in smoke; Dead against; Big gum; By the bye; At large.
3. (a) Make sentences with the following structures using the words given in brackets: (14)
 - i) Subject + Transitive Verb + Object + Noun complement (name as Verb)
 - ii) Subject + Transitive Verb + Indirect object + Direct object (write as Verb)
 - iii) There + Verb + subject (use any verb)
 - iv) Subject + Transitive Verb + Object (open us Verb)
 - v) Subject + Intransitive Verb + Adverbial complement (run as Verb)
 - vi) Subject + Transitive Verb + Object + Adjective (keep as Verb)
 - vii) Subject + Linking Verb + Noun complement (look as verb)
- (b) Frame sentences expressing the following attitudes/emotions. (12)
 - i) Command ii) Promise iii) Wish
 - iv) Greeting v) Regret vi) Apology
- (c) Supply a suitable word to fill the blanks in each of the following: (09)
 - i) I shall go out – it does not rain.
 - ii) They found the house –
 - iii) The teacher wants the notice –
 - iv) She like her tea –
 - v) He worked – the sun set.
 - vi) The team is as strong – its opponent.

4. (a) Correct the following sentences: (14)
- Kamal, you and I shall go to Dhaka.
 - I could not help to do this.
 - He discussed the matter in details.
 - You have played instead of worked.
 - It is a true fact.
 - My all books are lost.
 - Always speak truth.
- (b) Make sentences using the following modals as directed: (12)
- Should (to express duty)
 - Would (to express request)
 - Must (to express order)
 - May (to express permission)
 - Am + to (to express command)
 - Might (to express possibility)
- (c) Identify the parts of speech of the underlined words of the following sentences. (09)
- Our blessings come from above.
 - Analyse the above passage.
 - Draw near and listen to me.
 - His house is near the temple.
 - He is a near relation.
 - The time nears.

Section – B

5. (a) Read the following passage and answer the questions that follow: (20)
- The history of civilization shows how man always has to choose between making the right and wrong use of discoveries of science. This has never been more true than in our own age. In a brief period amazing discoveries have been made and applied to practical purpose. It would be ungrateful not to recognize how immense are the boons which science has given to mankind. It has brought within the reach multitudes of benefits and advantages which only a short time ago were the privileges of the few. It has shown how malnutrition, hunger and diseases can be overcome. It has not only lengthened life but it has deepened its quality. Fields of knowledge, experience and recreation open in the past only to a few have been thrown open to millions. Through the work of science the ordinary man today has been given the opportunity of a longer and fuller life than was ever possible to his grandparents. Most thankfully, we must acknowledge all that science has done and is doing for the welfare of our race.
- Questions:
- What does the history of civilization show?
 - Why is science important today?
 - How would man be ungrateful to science?
 - What welfare has science done for mankind?
- (b) Make a précis of the above passage (Q 5. a) with a suitable title. (15)
6. (a) Write a listing paragraph on postponement of examination. (15)
- (b) Amplify the idea contained in the following statement: (20)
- 'Man is born free but everywhere he is chained'.
7. (a) Write a report on "Celebration of University Day in Your Campus". (20)
- (b) Write a dialogue between two friends meeting unexpectedly in a railway carriage. (15)
8. Write a free composition on any one of the following. (35)
- Transport system of Bangladesh.
 - Global warming Bangladesh is a victim.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2014
Ch 1117
(Environmental Chemistry)

Full Marks: 210

Time: 3 hrs

- N.B.
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Necessary charts and graphs will be supplied

Section – A

1. (a) What is meant by corrosion? Discuss the factors that enhance the rate of corrosion in Khulna region. (09)
(b) Discuss the effect of P^H and electrolytes on the rate of underwater corrosion. (10)
(c) Write down the characteristic properties of industrial waste water. (08)
(d) Explain about homochain and heterochain polymers with examples. (08)
2. (a) What is radioactivity? Discuss about different types of radioactive radiation. (10)
(b) "Nuclear force is a special type of force"-Explain (08)
(c) Describe the mechanism of nuclear force. What is field particle? (12)
(d) The mass of ${}^{16}_8\text{O}$ is 15.994910 amu. Calculate its binding energy per nucleon. (05)
3. (a) What do you mean by the terms: (a) Substrate (b) Attacking reagent and (c) intermediates? Apply these terms to a substitution reaction. (10)
(b) What do you mean by Carbonium ions and Nucleophiles? Why tertiary Carbonium ions are more stable than primary Carbonium ions? (08)
(c) Discuss the mechanism of SN_1 and SN_2 reaction with suitable examples. (10)
(d) What do you mean by biodegradation of polymers? Why is it so important to the environment? (07)
4. (a) Describe the different transformation process of liquid pollutants. (10)
(b) Discuss about step reaction polymerization and free radical polymerization. (10)
(c) Describe the process of making polyethene from ethene by using Ziegler-Natta catalyst. (09)
(d) Explain H-bonding. (06)

Section – B

5. (a) Describe the trends in properties of group IA elements. (10)
(b) Explain why transition metal elements show variable oxidation state compare to alkali and alkaline earth metals. (10)
(c) Write down the possible reason for reactivity and softness character of group IA elements (10)
(d) Why does Fe^{2+} less stable than Fe^{3+} ? (05)
6. (a) What is the molecular orbital diagram for HeH^+ ? Discuss about stability of the ion? (09)
(b) Use molecular orbital theory to describe the bonding in O_2 . Find the bond order and decide whether it is paramagnetic or diamagnetic. (12)

- (c) Write a note on Bronsted Concept of acids and bases. (10)
 - (d) What are the conjugate bases of the following acids NH_3 , HClO_4 , HCN , H_2O (04)
7. (a) Discuss about the four types of environmental segments. (10)
- (b) What is acid rain? Discuss its harmful effect on environment. (10)
- (c) Explain the causes of depletion of Ozone layer. Why is ozone layer important for us? (10)
- (d) Write down the composition of atmosphere. (05)
8. (a) What do you understand by "Water quality parameters?" Explain the principle for measuring dissolved oxygen (DO) using winkler method. (10)
- (b) Write down the flow diagram for municipal sewage treatment. (09)
- (c) What is water pollution? Discuss about the various sources of water pollution. (10)
- (d) How does industrial waste pollute air? (06)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Examination, 2014

Math 1117
(Mathematics-I)

Full Marks. 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be supplied.

Section – A

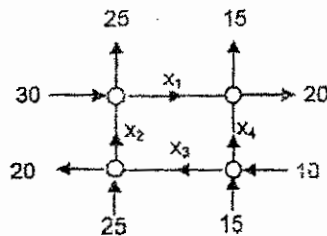
1. (a) If due to rotation of axes through an angle θ the equation $Ax^2+2Hxy+By^2=0$ becomes $ax'^2+By'^2=0$ then determine θ . (12)
- (b) What is the conic represented by $x^2-6xy+2y^2+10x-2y-5=0$? If the origin of the coordinates is shifted at the center of it, what will be the new form of its equation? (13)
- (c) What will be the new equation to the parabola $y^2=4ax$ after rotation of axis through an angle $\frac{\pi}{4}$. (10)
2. (a) Sketch the region common to $2x+3y \leq 6$, $4x+y \leq 4$, $x \geq 0$, $y \geq 0$, identifying the region in each case. (10)
- (b) With neat sketch obtain transformation equations among Cartesian, spherical polar and cylindrical polar coordinates. Obtain the spherical polar coordinates of $(-1, 1, -\sqrt{2})$. (25)
3. (a) Find the equation of the planes through $(0,4,-3)$, $(6,-4,3)$ which cuts off from the axes intercepts whose sum is zero. (10)
- (b) Define direction cosine and direction ratio. Let the direction ratios of two lines are given by $ul+vm+wn=0$ and $alm+bm+n+cnl=0$. Find the condition for which they will be perpendicular to each other. (14)
- (c) Find the distance of $(1,-1,1)$ from the plane $2x+3y+4z+5=0$ measured parallel to the line $x+y-z+3=0=3x+2y-2z-7$. (11)
4. (a) Let the plane $2x-3y+4z-12=0$ intersects the axes at A,B,C, planes through A,B,C parallel to coordinate planes meets at P. Find the centroid of the tetrahedron PABC. (10)
- (b) Find the Image of point (p,q,r) with respect to the plane $2x+y+z=6$. (10)
- (c) Let L and M be points on the lines $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{4} = \frac{z-7}{5}$ respectively such that LM is the shortest distance between the lines. Obtain the coordinates of L and M the equations to the line LM. (15)

Section – B

5. (a) Define Hermitian and skew Hermitian matrices. Express $\begin{bmatrix} 1+i & 0 & 2i \\ 3-2i & 2i & 5 \\ -5+3i & 3i & 7 \end{bmatrix}$ as the sum of a Hermitian and a skew Hermitian matrices. (14)
- (b) Define linear dependence and independence of vectors. Determine whether the following vectors in $\mathbb{R}^3$ are linearly dependent or not; $(1,-2,1)$, $(2,1,-1)$, $(7,-4,1)$. If they are dependent find a linear combination. (13)

- (c) For what values of a and b the matrix $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ will satisfy the equation (08)
 $A^2 + aA + bI = 0$, where I and 0 are the identity and null matrices respectively.

6. (a) Find the traffic flow in the adjoining network of one way streets. What may be the maximum traffic flow in the x_4 branch? (20)



- (b) Define singular matrix, upper triangular matrix and skew-symmetric matrix with example. (09)
 (c) Show that for a skew-symmetric matrix the diagonal elements are zero. (06)
7. (a) Define rank of a matrix. Determine the rank of the following matrices. (09)

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 & -1 \\ 3 & -2 & 1 & 4 & 7 \\ 4 & -1 & 3 & 7 & 6 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- (b) Test the consistency of the set of equations and if possible solve $y+z=1$, $2x-3y-z=5$, $4x+2y+z=3$, $6x+z=7$. (14)
 (c) Plot the following functions: $\frac{3}{y} = 2x-3$; $y = \frac{x^2+x-1}{x^2-7x+6}$; $y = 2\sin x + 3\sin 2x$ (12)
8. (a) Define the inverse of a square matrix. Test the singularity of (15)

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 3 & -1 \\ -1 & -2 & 7 \end{bmatrix} \text{ and if possible find the inverse of } A.$$

- (b) A bike costs Tk. 2/km when it is at the speed of 25km/hr and Tk. 5/km at the speed of 40km/hr. If a rider wants to spend a maximum of Tk. 100 for riding, what the maximum distance can be travelled by the rider within an hour? (20)

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 1st Year 1st Term Examination, 2014
URP 1113
(Fundamentals of Planning Process)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) As a student of Urban and Regional Planning, what is your understanding about "Planning" and the activities of a "Planner"? (10)
(b) Write down the importance of studying fundamentals of planning process for a planner. (10)
(c) Briefly discuss with example the stages of planning process by John Ratcliffe. (15)
2. (a) Define Master Plan. (05)
(b) Show the process of preparation of Khulna City Master Plan 2001 in a schematic diagram. (15)
(c) How will you differentiate the following forms of planning-"Local Planning", "Regional Planning" and "National Planning". Use appropriate example. (15)
3. (a) Briefly elaborate the simplified series of steps followed in the formulation of "Annual Development Programme". (05)
(b) Discuss the important targets set in the SFYP for the progress and development of Bangladesh. (10)
(c) Write short notes on the followings (any four): (4x5=20)
 - i) Stages of town development by Lewis Mumford
 - ii) Classification of cities in terms of population
 - iii) Dimensions of Planning
 - iv) Physical and topographic survey
 - v) Urbanization of Bangladesh and UDD
 - vi) Development Authorities in planning and development control.
4. (a) Make a comparative analysis between "PRSP" and "Mid Term (5 year) Plan". (05)
(b) Mention the structure of the Planning Commission of Bangladesh. (08)
(c) Write down the functions of the General Economics Division and Physical Infrastructure Division. (07)
(d) Show the framework of the steps in processing and approving the investment projects. (10)
(e) Write a short note on Perspective Plans in Bangladesh. (05)

Section – B

5. (a) Elaborate the chronological change in the profession of planner from designer to negotiator. (15)
- (b) Write about the principles of planning theory. How can you apply planning theory in the practical field of planning? (10)
- (c) Make a comparison between "Positivism" and "Post-positivism" approach of planning theory. Use necessary example. (10)
6. (a) What is your understanding about "Comprehensive Plan"? Discuss about the planning process followed in "Comprehensive Plan" with an example. (15)
- (b) Identify some criticisms of "Comprehensive Plan". (05)
- (c) Suppose, you are facing a problem in your real life. How will you solve this problem following the approach of "Rational Planning"? Describe with necessary example. (10)
- (d) In "Advocacy Planning, planners play the role of pleader"-explain this statement. (05)
7. (a) "Transactive planning is a participatory planning approach"-discuss about the role of planner and general people in this planning process. (10)
- (b) In which situation "Incremental Planning" is more applicable? What are the important features of this planning? (15)
- (c) Discuss about the three conflicting planning goals balanced through "Adaptive Sustainable Planning". (05)
- (d) What are the major principles of "Radical Planning"? (05)
8. (a) Briefly discuss the sustainability triangle of Sustainability Planning Model with a framework. (15)
- (b) Write short notes on the following issues: (any four) (4x5=20)
- i) Decentralization
 - ii) Land use zoning
 - iii) Constructivism
 - iv) Soft facts in planning
 - v) Equity vi) Mutual learning.
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